SHIVSHANKER PATHAK

Application Architect

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Application Architect (Java & Cloud-Native) Performance-focused Architect with 13+ years of experience designing scalable, distributed systems for the Fintech and Retail Banking sectors. Deep expertise in building low-latency payment rails (NEFT/RTGS) and event-driven architectures. Specialist in **Spring Boot 3**, **Distributed Caching**, **Microservices** and **AWS Services**. Committed to writing clean, maintainable code and implementing robust CI/CD pipelines to support high-frequency deployments in mission-critical environments.

Professional Summary

- Senior Application Architect with extensive experience in the design, development, and implementation of enterprise-grade applications. Expert in **Java (8/17/21)**, **Spring Boot 3+, and Spring Cloud**, with a deep focus on **Microservices architecture**. Proven track record of building scalable, cloud-native solutions and optimizing back-end performance to meet complex business requirements.
- Strong knowledge in **Java multithreading, executer framework, concurrent collections, virtual threads.**
- Proven track record of designing, implementing and integration of software systems with **Java (8,17 & 21), Spring Boot 3+, Spring Cloud, Spring Security, Hazel cast, Apache Kafka & Redis.**
- Extensive hands-on experience in relational databases **(Oracle, MySQL)** and NoSQL Databases **(MongoDB, CouchDB)**, including schema design, query optimization, indexing, and performance tuning.
- Solid understanding of cloud computing and practical knowledge of key **AWS services including EC2, VPC, S3, RDS, AWS Lambda & API Gateway.**
- Expertise in Software Architecture: Strong command of **SOLID** principles and **Design Patterns.**
- **Extensive hands-on experience architecting Event-Driven Microservices using Circuit Breakers** (Resilience4j), API Gateways, and service discovery to build resilient, fault tolerant distributed systems.
- Comprehensive understanding of payment systems including **UPI, NEFT, RTGS, and digital banking platforms.**
- **Strong advocate of Continuous Integration and Continuous Deployment** (CI/CD) practices, leveraging containerization with **Docker** and **Kubernetes** for deployments, and utilizing tools such as **Jenkins** and **SonarQube** to maintain code quality and accelerate release cycles.
- **Led and mentored a cross-functional** team of 15+ members, overseeing end-to-end project management and the delivery of high-concurrency Java applications.
- Hands of experience for Observability tool such **Dynatrace**, **Grafana Prometheus**

Work Experience

Application Architect Mar 2024 - Present

IBM India Pvt Ltd. (Mumbai)

Define the architecture of scalable, high-performance Java applications, ensuring they meet business requirements and technical specifications.
Full ownership of project deliverables, leading team, collaboration and mentoring.
Co-ordination with solution architect, project manager and client to complete the change request as per timeline, estimate delivery timeline.
Interact with information security team to address vulnerability point and mitigate roadblock for production signoff.
Provide post deployment production support and monitoring the system.

- Led the development and implementation of scalable, high-performance Java-based modules for the SBI Yono mobile application.
- Architected the migration of session data from **IBM Dynacache** to a **Hazelcast-based** distributed caching solution.
- Designed and implemented microservices-based architectures, introduce Spring Boot 3 light weight APIs and decommission Monolith MFP framework.
- Engineered high-performance mobile backend features by leveraging **Spring Boot and Spring Cloud** for microservices orchestration, utilizing **Redis Cache** to reduce latency and **MongoDB** for resilient data persistence.

Achievements:

Successfully delivered the MFP decommission and migrated 1,200+ APIs to Spring Boot.
Transition of SBI YONO session management to Hazelcast IMDB Grid.

Technical Lead Oct 2018 - Mar 2024

IBM India Pvt Ltd. (Mumbai)

Technical Skills

Java (17&21)	●●●●●
Spring Boot 3.2+	●●●●●
Spring Cloud	●●●●●
Microservices	●●●●●
Oracle	●●●●◐
MongoDB	●●●●◐
Redis	●●●●◐
Hazelcast	●●●●◐
Java Multithreading	●●●●◐
Design Patterns	●●●●◐
SOLID Principles	●●●●●
CI-CD	●●●●◐
AWS	●●●●◐
Kubernetes	●●●●◐
Docker	●●●●◐
JavaScript	●●●◐○
Apache Kafka	●●●●◐
IBW WAS & Liberty	●●●●◐
Linux	●●●●◐

Soft Skills

Communication
Teamwork
Leadership
Problem-solving

Core Competencies

Java Microservices
Software Developement
Design Principles
Software Architecture

Certifications

- **Executive Post Graduate Certification in Cloud Computing & DevOps, iHUB Divya Sampark.** Indian Institute of Technology Roorkee in association with Intellipaat. (2025)

Mentored and led a cross-functional team of 10+ developers through the full SDLC, fostering a culture of continuous learning and technical excellence.
Provided technical leadership across architecture, design, development, and production support for mission-critical systems.
Owned end-to-end delivery, including requirement analysis, solution design, implementation, and release management.
Ensured high code quality through CI/CD pipelines, automated testing, and secure coding standards.

- **Created backend functionality for Mobile app** from scratch using Spring Boot microservices, DB2, and IBM WebSphere Application.
- Developed key features and module on SBI Yono App like **IRCTC** Integration, **OMP**, **PFM** Systems and delivered 50+ new and enhancement projects.
- **Enhanced overall app & API performance** using API caching, code refactoring, database query optimization and leverage Jave 17 & 21 feature,
- Outline the system architecture to integrate microservices and third-party applications using **RESTful** web services.
- Built Notification engine for SBI YONO Mobile App which covered SMS, Email and Push Notification.
- Mentored junior team members and conducted technical interviews for project staffing.

Senior Software Engineer Jan 2016 - Oct 2018
IDBI Intech Ltd. (Navi Mumbai)

Led development and enhancement of key modules in the IDBI Payments ecosystem to improve reliability and performance.
Provided L2/L3 production support for NEFT/RTGS platforms, ensuring high availability and quick resolution of critical issues.
Designed and implemented multithreaded background services for NEFT & RTGS to support high-volume transaction processing and faster settlement cycles.
Modernized the NEFT application by migrating from JSP-Servlet to Spring MVC, improving maintainability and development efficiency.

- Developed key payment applications such as NEFT and RTGS for IDBI Bank's for IDBI Branch operations and customers.
- Designed and implemented the NEFT module for IDBI Bank branches using Core Java, Spring Boot, AngularJS, and Oracle database.
- Provided L2/L3 production support for enterprise banking payment platforms including NEFT, RTGS, and CPS systems, ensuring uninterrupted services and quick issue resolution.
- Built responsive and user-friendly interfaces using Spring MVC, JavaScript, and Bootstrap, enhancing user experience and operational efficiency.
- Developed multithreaded background services in Core Java with Java Executer framework to process bulk financial transactions, improving throughput and reducing processing time.
- Created and optimized Oracle PL/SQL procedures, functions, and scripts to support secure, scalable backend transaction processing.

Software Engineer Aug 2012 - Jan 2016
Alepo Technologies (Navi Mumbai)

Developed and enhanced multiple features of the Liferay portal, including User Management, Audit Trail, and Voucher Management modules.
Provided production support and resolved defects for the CRM Portal, ensuring smooth portal operations and improved user experience.
Collaborated with business and support teams to analyze issues, apply fixes, and deliver portal enhancements in timely manner.

- Built and enhanced Liferay portal features including User Management, Audit Trail, and Voucher Management System.
- Operated as a Java Portlet Developer to construct CRM features aligned with business and compliance requirements.
- Resolved production issues, performed bug fixes, and delivered ongoing technical support for the CRM Portal, ensuring smooth operations.

Education

Post Graduate Diploma in Software Technolgy (2011 - 2012)
CDAC Kharghar, CDAC Kharghar Navi Mumbai

Bachelor of Science in Information Technology (2006 - 2009)
Sikkim Manipal University

Achievements

- Awarded Ace Employee of the Year 2022 by IBM India Pvt Ltd for outstanding performance and contributions to the YONO App development.
- Certificate of Appreciation from SBI Client for building the entire backend system for YFEI Mobile App.
- Certificate of Appreciation from SBI Client for successful migration of YONO system to Hazelcast caching.

Projects

Migration of MobileFirst Adapters to IBM API Connect with Spring Boot Microservice
Project Description
Migration of IBM MobileFirst Adapters to Spring Boot microservices to improve performance, scalability and monetization of the YONO Business APIs.
Roles & Responsibility
- Create high-level and low-level architecture design, take design decision for overall migration flow.
- Led the design and implementation of a scalable backend platform, including APIC OAuth2 flow integration, end-to-end migration of IBM MobileFirst Adapter APIs to Spring Boot microservices with improved performance, maintainability, and security.
- Drive the complete SDLC life cycle, provide technical support, bug fixing.
- Lead the small team of developers, assign task and monitor the progress of project.
- Created comprehensive technical documentation such as HLD, Sequence flow.
- Design review, prepare deployment strategy and monitor every stage for software development

Technology Stack
Java 17, Kafka, Spring Boot 3.2, Spring Cloud, IBM API Connect, Jenkins, Maven, Resilience4j, Mongo DB, DB2, Jenkins, IBM Liberty, Redis, Hazelcast.

Integration of Hazelcast Distributing Caching in YONO
Project Description

Migration of YONO mobile app session management from IBM WAS Dyn cache to Hazel cast IMDG to improve system performance and achieve high number of user login.

Roles & Responsibilities

- Design the complete system architecture of Hazel cast cluster in Yono system, contribute to various design decision, capacity planning, client configuration.
- Configuration & installation of Hazel cast cluster and management console in staging and production system.
- Configure Yono's backend applications to use Hazel cast clients for session data, OTP caching, and synchronization.

Technology Stack

Java 17, Hazelcast, Spring Boot, Spring Cloud, Mongo DB

Yono for Every Indian UPI integration.

Project Description

YFEI is a SBI Mobile Application for UPI transaction from any RBI registered bank of India.

Roles & Responsibilities

- Design and implement system architectures, solutioned low-level design (LLD) and high-level design (HLD) for YFEI system integration with UPI.
- Outlined the complete system flow between SBI mobile app, SDK and Yono blackened system.
- Developed entire backend systems, microservices, integration with NPCI, session management.
- Development of features like user Login and Registration, user profile management, integration with ELK system for auditing and logging.
- Integration with Hazel cast for session management.
- Take complete ownership for deployment in production system with Jenkins.
- Implement security feature and recommendation as per ISD department.

Technology Stack

Java 21, IBM Liberty, Spring Boot 3.2, DB2, Apache Kafka, Event driven microservices, ELK

Notification Engine for SBI YONO

Project Description

Custom notification platform to trigger targeted, event base and bulk notification for SMS, Email and Push notification.

Roles & Responsibilities

- Owned the end-to-end architecture of a scalable, multi-channel notification system, translating business requirements into resilient, extensible system design and guiding implementation across teams.
- Delivered event-driven microservices for Push and Email notifications, defining robust API contracts, security protocols, and seamless integrations with external systems including SMS gateways, Google Firebase (FCM), and enterprise SMTP servers.

Technology Stack

Java 17, IBM Liberty, Spring Boot 3.2, Resilience4j, DB2, Apache Kafka, Event driven microservices, ELK, Jenkins, OpenShift, Padman, OAuth2, IBM APIC Connect.

Hobbies

Exploring new technology trends, Playing Cricket, Reading Books

Languages

Hindi, English